

ALSHUMOOKH GROUP

Settlement Gateway v2.0 — Technical Integration Division

TECHNICAL INTEGRATION
ROADMAP & API SPECIFICATION

HIPERCAPITAL FINANCE ↔ ALSHUMOOKH SETTLEMENT GATEWAY v2.0

VERSION 3.0 — FULL TECHNICAL SPECIFICATION

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■ IMPORTANT NOTICE:

This document contains proprietary API specifications and live system details of ALSHUMOOKH GROUP settlement infrastructure. Unauthorized distribution is strictly prohibited. All integration attempts are logged and monitored.

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1. EXECUTIVE SUMMARY & INTEGRATION OVERVIEW

ALSHUMOOKH GROUP operates the Alshumookh Settlement Gateway v2.0 (ASIG), a production-grade blockchain settlement infrastructure built on FastAPI. This document provides HIPERCAPITAL FINANCE with the complete technical specification required to integrate their HIPCF Chain (Chain ID 240884) with ASIG. Every field alias, status code, error code, authentication method, and API endpoint documented here reflects the live production system.

Component	ASIG Details	HIPCF Requirement
Gateway	ALSHUMOOKH Settlement Gateway v2.0	Must integrate via REST API
Framework	FastAPI (Python 3.11+)	Standard HTTPS REST
Accepted Assets	USDT (ERC-20/TRC-20/Base), USDC, M1 Tokens	HCEURO must bridge to USDT first
Settlement Networks	Ethereum, Base, Tron (placeholder)	HIPCF Chain NOT supported natively
Blockchain Verification	Alchemy Webhooks + RPC	tx_hash on public chain required
Auth Methods	API Key, OAuth2, HMAC-SHA256, JWS, JWE, mTLS	Minimum: API Key + HMAC-SHA256
Rate Limit	240 req/60s global — 200 req/60s per API key	Must implement request queuing

2. SYSTEM ARCHITECTURE & SECURITY REQUIREMENTS

The ASIG pipeline processes every inbound settlement payload through a mandatory multi-stage security and verification pipeline. HIPERCAPITAL FINANCE must ensure their integration complies with each layer.

■ PIPELINE FLOW: Inbound Request → IP Allowlist Check → API Key / OAuth2 Authentication → Idempotency Check (Idempotency-Key header) → mTLS Certificate Verification (if required) → JWE Decryption (if encrypted) → JWS Signature Verification → HMAC-SHA256 Validation → JSON Normalization & Field Alias Resolution → Database Storage → Alchemy Blockchain Verification → On-Chain Confirmation → Reconciliation

Security Layer Requirements

Security Layer	Header / Mechanism	Required?	Details
IP Allowlist	Server-side check	✓ Required	HIPCF integration IP(s) must be registered in ASIG dashboard
API Key	X-API-Key:	✓ Required	Generated from ALSHUMOOKH admin dashboard
Idempotency	Idempotency-Key:	✓ Required	Unique UUID per request — prevents duplicate processing
OAuth2 Bearer	Authorization: Bearer	Optional	Required if client config sets require_oauth=True
HMAC-SHA256	X-HMAC-Signature:	Recommended	HMAC of: METHOD + URL + timestamp + body_hash using shared secret
Timestamp	X-Timestamp:	With HMAC	Unix timestamp — max 5 minutes drift allowed
JWS Signing	X-JWS-Signature:	Optional	JSON Web Signature of full request body
JWE Encryption	Content-Type: application/jose	Optional	Full body encrypted as JWE compact serialization
mTLS	X-Client-Cert-Fingerprint:	Optional	Mutual TLS — client certificate SHA-256 fingerprint

3. AUTHENTICATION METHODS — COMPLETE SPECIFICATION

3.1 — API Key Authentication (Minimum Required)

Every request to ASIG must include the X-API-Key header. Keys are generated from the ALSHUMOOKH admin dashboard and are hashed with SHA-256 before storage. An inactive or invalid key returns `error_code: invalid_api_key` (HTTP 401).

Request Header:

```
X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000
```

3.2 — OAuth2 Client Credentials Flow

For clients requiring token-based authentication, ASIG supports OAuth2 Client Credentials. POST to `/api/v1/oauth/token` with `client_id` and `client_secret` to obtain a Bearer token (expires in 3600s). Use the token in subsequent requests.

Token Request:

```
POST /api/v1/oauth/token
Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials
&client_id=<your_client_id>
&client_secret=<your_client_secret>

Response:
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in": 3600
}
```

Use in requests:

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

3.3 — HMAC-SHA256 Request Signing (Recommended)

Recommended for all production integrations. The HMAC secret is assigned per API client and stored in the ASIG database. The signature covers the HTTP method, URL, Unix timestamp, and SHA-256 hash of the request body.

Signature Construction:

```
# Python example:
import hmac, hashlib, time, json

method = 'POST'
url = '/api/v1/payloads/ingest'
timestamp = str(int(time.time())) # Unix timestamp
body = json.dumps(payload) # JSON string
body_hash = hashlib.sha256(body.encode()).hexdigest()

message = f'{method}\n{url}\n{timestamp}\n{body_hash}'
signature = hmac.new(secret.encode(), message.encode(),
hashlib.sha256).hexdigest()

Headers:
X-HMAC-Signature: <hex_signature>
X-Timestamp: <unix_timestamp> # Must be within 5 minutes
```

4. MAIN INGEST API ENDPOINT — FULL TECHNICAL DETAILS

Property	Value
Endpoint	POST /api/v1/payloads/ingest
Content-Type	application/json (or application/jose for JWE)
Auth Required	X-API-Key (+ Idempotency-Key header)
Idempotency	Idempotency-Key: – REQUIRED on every request
Rate Limit	200 req / 60s per API key 240 req / 60s global
Success Response	HTTP 200 {status: payload_received, payload_id: ..., verification_status: ...}
Duplicate Key	HTTP 200 {status: payload_received (idempotency replay), parsed: true/false}
Auth Failure	HTTP 401 / 403 – See error codes in Section 7
Validation Fail	HTTP 400 – missing field or invalid format

4.1 — Complete cURL Request Example

```
curl -X POST https://api.alsumookh.com/api/v1/payloads/ingest \
-H 'Content-Type: application/json' \
-H 'X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX' \
-H 'Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000' \
-H 'X-HMAC-Signature: a3f5c8d2e9b1...' \
-H 'X-Timestamp: 1749461234' \
-d '{
  "transaction_reference": "HIPC-2026-001",
  "tx_hash": "0xabc123...def456",
  "sender_wallet": "0xSENDER_WALLET_ADDRESS",
  "receiver_wallet": "0xBD682cfD8382a90adFd6745780D3D7959c4d939",
  "amount": "50000.00",
  "asset": "USDT",
  "network": "ethereum",
  "token_contract": "0xdAC17F958D2ee523a2206206994597C13D831ec7",
  "settlement_type": "wallet_to_wallet",
  "callback_url": "https://your-server.com/webhook/asig",
  "timestamp": "2026-06-09T10:00:00Z"
}'
```

4.2 — Successful Response Examples

Response (with tx_hash — goes to Alchemy verification):

```
HTTP 200 OK
{
  "status": "payload_received",
  "payload_id": "a8f2c3d4-1234-5678-abcd-ef0123456789",
  "transaction_reference": "HIPC-2026-001",
  "parsed": true,
  "verification_status": "ALCHEMY_PENDING",
  "network": "ethereum",
  "amount": "50000.00",
  "asset": "USDT",
  "tx_hash": "0xabc123...def456",
  "idempotency_key": "550e8400-e29b-41d4-a716-446655440000",
  "created_at": "2026-06-09T10:00:05Z"
}
```

Response (without tx_hash — awaiting transaction hash):

```
HTTP 200 OK
{
  "status": "payload_received",
```

```
"payload_id": "b9e3d4f5-2345-6789-bcde-f01234567890",  
"parsed": true,  
"verification_status": "AWAITING_TX_HASH",  
"message": "Payload received. Submit tx_hash when transaction is broadcast."  
}
```

Response (parse failed — body not JSON or unrecognized structure):

```
HTTP 200 OK  
{  
  "status": "payload_received_unparsed",  
  "payload_id": "c0f4e5a6-3456-789a-cdef-012345678901",  
  "parsed": false,  
  "verification_status": "RECEIVED",  
  "message": "Payload stored but could not be normalized. Manual review required."  
}
```

5. COMPLETE JSON PAYLOAD SCHEMA & FIELD ALIASES

ASIG's normalization engine resolves incoming JSON fields using a comprehensive alias mapping. This means HIPERCAPITAL FINANCE can use their own field names — the system will auto-detect and map them to the canonical ASIG field. The table below shows every canonical field, its accepted aliases, type, and requirement level.

Canonical Field	Accepted Aliases (all resolve to canonical)	Type	Required?
transaction_reference	transaction_id · txRef · reference · ref · transactionCode · referenceld · payment_reference	string	Recommended
tx_hash	transaction_hash · hash · blockchain_hash · txid · txHash · transactionHash	string (0x...)	Required if sending
sender_wallet	from_wallet · from · source_wallet · origin_wallet · fromAddress · from_address · senderAddress	string (0x... or T...)	Required
receiver_wallet	to_wallet · to · destination_wallet · beneficiary_wallet · toAddress · to_address · recipientAddress	string (0x... or T...)	Required
amount	value · token_amount · transfer_amount · settlement_amount · tokenAmount · transferAmount	decimal string	Required
asset	currency · token · symbol · crypto_currency · tokenSymbol · cryptoCurrency	string (USDT/USDC)	Required
network	chain · blockchain · protocol · chainName · networkName	string (ethereum/base/tron)	Required
token_contract	contract_address · contract · token_address · contractAddress · tokenContract	string (0x...)	Recommended
timestamp	created_at · time · tx_time · txTime · transaction_time	ISO8601 / Unix	Optional
sender	sender_name · from_name · senderName · fromName · payer	string	Optional
receiver	receiver_name · to_name · beneficiary · receiverName · toName · payee	string	Optional
settlement_type	settlement · type · settlementType · paymentType · payment_type	string	Optional
authorization_code	auth_code · authorization · authCode · authorizationCode	string	Optional
callback_url	callback · webhook_url · notify_url · callbackUrl · webhookUrl · notifyUrl	URL string	Optional
signature	sig · x_signature · xSignature · payloadSignature	hex string	Optional
payload_hash	hash_payload · content_hash · payloadHash · contentHash	SHA-256 hex	Optional

5.1 — Complete JSON Schema (all fields)

```
{
  "transaction_reference": "HIPC-2026-001", // string – your internal TX ref
  "tx_hash": "0xabc123...def456", // EVM tx hash (0x...) or TRON hash
  "sender_wallet": "0xYOUR_WALLET", // source address (EVM: 0x, Tron: T...)
  "receiver_wallet": "0xBD682cfD8382a90adFd6745780D3D7959c4d939", // ASIG M1 wallet
  "amount": "50000.00", // decimal string – USDT amount
  "asset": "USDT", // supported: USDT, USDC, M1
  "network": "ethereum", // ethereum | base | tron
  "token_contract": "0xdAC17F958D2ee523a2206206994597C13D831ec7", // USDT ERC-20
  "settlement_type": "wallet_to_wallet", // optional context field
  "timestamp": "2026-06-09T10:00:00Z", // ISO8601
  "sender": "HIPERCAPITAL FINANCE", // optional sender name
  "receiver": "ALSHUMOOKH GROUP", // optional receiver name
  "authorization_code": "HIPC-AUTH-2026", // optional authorization ref
}
```

```
"callback_url": "https://your-server.com/webhook/asig" // optional
}
```

✗ IMPORTANT: The system does NOT accept: asset=TOKENIZED_USD, asset=USDx, asset=HCEURO, settlement_type=PRIVATE_INTERNAL_LEDGER, or any non-blockchain settlement mode. HIPERCAPITAL must bridge HCEURO to USDT on a public EVM chain before submitting a payload.

6. PAYLOAD VERIFICATION STATUS FLOW

Every payload submitted to ASIG transitions through a defined verification state machine. HIPERCAPITAL FINANCE should poll GET /api/v1/payloads/{payload_id} or use the callback_url webhook to receive real-time status updates.

Status	Meaning	Next Step	Action Required
RECEIVED	Payload stored but parsing failed / unrecognized structure	Manual review	Contact ASIG operations — check your JSON structure
PARSED	Fields normalized and extracted successfully	→ AWAITING_TX_HASH or ALCHEMY_PENDING	Provide tx_hash if not included
AWAITING_TX_HASH	Payload parsed but no blockchain tx_hash provided	Submit tx_hash	PATCH /api/v1/payloads/{id} with tx_hash once broadcast
ALCHEMY_PENDING	tx_hash submitted — awaiting Alchemy blockchain confirmation	Automatic (webhook)	No action — wait for blockchain confirmations
ALCHEMY_VERIFIED	Alchemy confirmed tx on-chain (6+ blocks ETH/Base, 20 TRON)	→ ON_CHAIN_CONFIRMED	System auto-advances — no action needed
ON_CHAIN_CONFIRMED	Transaction fully confirmed on blockchain	→ RECONCILED	ASIG operations reconciles and releases settlement
RECONCILED	Settlement complete — funds reconciled in ASIG ledger	Terminal state ✓	Settlement process complete
MANUAL_REVIEW	Flagged for human review (RPC error, TRON, mismatched amount)	ASIG ops review	Contact ASIG operations with payload_id
FAILED	Blockchain verification failed or amount mismatch detected	Terminal state ✗	Resubmit with corrected data or contact ASIG ops

6.1 — Outbound Transfer Status (ASIG → HIPERCAPITAL)

Status	Meaning
PENDING	Transfer created, awaiting processing
AWAITING_APPROVAL	Requires admin approval before broadcast
APPROVED	Approved — queued for blockchain broadcast
BROADCASTING	Transaction being broadcast to network
COMPLETED	Transaction confirmed on-chain — settlement complete
FAILED	Broadcast failed — retry or manual intervention required
CANCELLED	Transfer cancelled before broadcast

7. ALL ERROR CODES & REJECTION REASONS

All error responses from ASIG follow a consistent JSON structure. The `error_code` field uniquely identifies the rejection reason. Below is the complete list of all error codes, their HTTP status, cause, and corrective action.

Error Response Structure:

```
HTTP 4xx / 5xx
{
  "error": "error_code_string",
  "message": "Human-readable description of the rejection reason"
}
```

HTTP	error_code	Event Logged	Cause	Corrective Action
401	missing_credentials	PAYLOAD_REJECTED_AUTH	No X-API-Key and no Authorization header	Add X-API-Key or OAuth Bearer token
401	invalid_api_key	PAYLOAD_REJECTED_AUTH	X-API-Key does not match any active client	Verify API key from ALSHUMOOKH dashboard
401	oauth_required	PAYLOAD_REJECTED_AUTH	Client config requires OAuth2 but API Key sent	Obtain Bearer token via /api/v1/oauth/token
403	ip_not_allowed	PAYLOAD_REJECTED_AUTH	Caller IP not in client's IP allowlist	Register your egress IP in ASIG dashboard
400	missing_idempotency_key	PAYLOAD_REJECTED_VALIDATION	Idempotency-Key header absent	Add Idempotency-Key: to every request
401	mtls_required	PAYLOAD_REJECTED_AUTH	Client requires mTLS but no cert fingerprint	Add X-Client-Cert-Fingerprint header
400	invalid_jwe	PAYLOAD_REJECTED_VALIDATION	JWE-encrypted body could not be decrypted	Check JWE key pair and encryption parameters
401	timestamp_expired	PAYLOAD_REJECTED_AUTH	X-Timestamp is older than 5 minutes	Sync your server clock (NTP) — use UTC
401	invalid_jws	PAYLOAD_REJECTED_AUTH	JWS signature on body does not verify	Check your signing key and JWS algorithm
401	hmac_required	PAYLOAD_REJECTED_AUTH	Client config requires HMAC but none provided	Add X-HMAC-Signature + X-Timestamp headers
401	invalid_hmac	PAYLOAD_REJECTED_AUTH	HMAC-SHA256 signature verification failed	Check HMAC secret and message construction
400	invalid_json	N/A	Request body is not valid JSON	Ensure Content-Type: application/json
400	invalid_payload	N/A	JSON body is not a JSON object (e.g. array)	Body must be a JSON object {}
400	missing_api_key	N/A	M1 Fund endpoint: X-API-Key absent	Add X-API-Key header to M1 fund requests
404	Payload not found	N/A	GET /payloads/{id} — ID does not exist	Check payload_id from ingest response
429	rate_limit_exceeded	N/A	Exceeded 200 req/60s per key or 240/60s global	Implement exponential backoff and request queue

8. ALL API ENDPOINTS REFERENCE

8.1 — Settlement / Payload Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/payloads/ingest	API Key + Idempotency-Key	Submit settlement payload — main integration endpoint
GET	/api/v1/payloads/schema	API Key	Retrieve canonical schema definition and supported fields
POST	/api/v1/payloads/m1-fund	API Key	M1 Fund inbound settlement (EUR wire / IBAN pathway)
GET	/api/v1/payloads/{id}	API Key	Retrieve payload status and details by payload_id
POST	/api/v1/oauth/token	Client credentials	Obtain OAuth2 Bearer token (expires 3600s)

8.2 — Payment Order Endpoints

Method	Endpoint	Description
POST	/api/v1/payments/orders	Create payment order (Coinbase/MoonPay/Circle/Stripe)
GET	/api/v1/payments/orders	List all payment orders
GET	/api/v1/payments/orders/{id}	Get payment order by ID
GET	/api/v1/payments/orders/{id}/status	Get order status
GET	/api/v1/payments/widget-url	Get onramp widget URL
GET	/api/v1/payments/coinbase/onramp-url	Get Coinbase onramp URL
GET	/api/v1/payments/transactions	List all transactions
GET	/api/v1/payments/transactions/{id}	Get transaction by ID

8.3 — Webhook Receiver Endpoints

Method	Endpoint	Source	Description
POST	/webhooks/alchemy	Alchemy	Blockchain tx confirmations — auto-advances payload status
POST	/webhooks/circle	Circle	Circle payment confirmations
POST	/webhooks/moonpay	MoonPay	MoonPay payment events
POST	/webhooks/coinbase	Coinbase	Coinbase Commerce events

8.4 — Admin Endpoints (Internal Use / Authorized Partners)

Method	Endpoint	Description
GET	/api/v1/admin/payloads	List all payloads with filters (status, date, network)
GET	/api/v1/admin/payloads/{id}	Get full payload detail (admin view)
POST	/api/v1/admin/payloads/{id}/verify	Trigger manual blockchain re-verification
PATCH	/api/v1/admin/payloads/{id}/review	Update manual review decision
POST	/api/v1/admin/outbound-transfers	Initiate USDT outbound transfer (ETH/TRON/Base)

GET	/api/v1/admin/outbound-transfers	List outbound transfers with status
GET	/api/v1/admin/clients	List API clients
POST	/api/v1/admin/clients	Create new API client
GET	/api/v1/admin/dashboard/stats	System statistics and payload counts
GET	/api/v1/admin/treasury/balance	Treasury wallet balances
POST	/api/v1/admin/fnfcu	FNFCU transfer request submission

9. BLOCKCHAIN NETWORKS & CONTRACT ADDRESSES

The following contract addresses are live production values. All settlements must be sent to the correct contract and receiver wallet for the respective network. Verification thresholds are enforced by the Alchemy webhook integration.

9.1 — Supported Token Contracts

Token	Network	Contract Address	Standard	Status
USDT	Ethereum Mainnet	0xdAC17F958D2ee523a2206206994597C13D831ec7	ERC-20	✓ Active
USDT	Tron	TR7NHqjeKQxGTCi8q8ZY4pL8otSzgJLj6t	TRC-20	■ Placeholder
USDC	Ethereum Mainnet	0xa0b86991c6218b36c1d19d4a2e9eb0ce3606eb48	ERC-20	✓ Active
USDC	Base Network	0x833589fcd6edb6e08f4c7c32d4f71b54bda02913	ERC-20 (Base)	✓ Active
M1 Token	Sepolia Testnet	0xD999DB972BBDc2C13a9595A1474A04F5e59169a5	ERC-20	Testnet Only
SIG Token	Sepolia Testnet	0x14E068D3C99cbC8BAe723D83D1761EF4C124A0ab	ERC-20	Testnet Only

9.2 — Blockchain Confirmation Thresholds

Network	Chain ID	Min Confirmations	Avg Time to Confirm	Verification Method
Ethereum Mainnet	1	6+ blocks	~90 seconds	Alchemy Webhook + RPC
Base Network	8453	6+ blocks	~18 seconds	Alchemy Webhook + RPC
Tron	N/A	20 blocks	~60 seconds	MANUAL_REVIEW (placeholder)
Sepolia Testnet	11155111	6+ blocks	~90 seconds	Alchemy Webhook (testing)

9.3 — ASIG Receiver Wallet Addresses

- All USDT/USDC settlements must be sent to the designated ASIG receiver wallets below. The receiver_wallet field in your payload must match these addresses exactly. Sending to any other address will result in MANUAL_REVIEW status.

Network	ASIG Receiver Wallet / Account	Asset
Ethereum / Base	0xBD682cfD8382a90adFdD6745780D3D7959c4d939	USDT / USDC (ERC-20)
Tron (TRC-20)	Contact ASIG ops for TRON address	USDT (TRC-20)
Wire Transfer (IBAN)	AE960260001015754367301	USD / EUR (Fiat)
Bank Name	Emirates NBD — Sheikh Zayed Road, Dubai, UAE	—
BIC / SWIFT	EBILAEADXXX	—

10. WEBHOOK & CALLBACK SPECIFICATION

When a payload's verification_status changes, ASIG posts a JSON notification to the callback_url provided in the original payload. HIPERCAPITAL FINANCE must implement a publicly accessible HTTPS endpoint to receive these updates.

10.1 — Outbound Callback Payload (ASIG → HIPERCAPITAL)

```
POST https://your-server.com/webhook/asig
Content-Type: application/json

{
  "event": "payload_status_update",
  "payload_id": "a8f2c3d4-1234-5678-abcd-ef0123456789",
  "transaction_reference": "HIPC-2026-001",
  "verification_status": "ON_CHAIN_CONFIRMED",
  "previous_status": "ALCHEMY_VERIFIED",
  "tx_hash": "0xabc123...def456",
  "network": "ethereum",
  "amount": "50000.00",
  "asset": "USDT",
  "timestamp": "2026-06-09T10:05:22Z"
}
```

10.2 — Callback Security Requirements

Your webhook endpoint must respond with HTTP 200 within 10 seconds. ASIG will retry failed deliveries up to 3 times with exponential backoff. Validate the callback using the HMAC-SHA256 signature in the X-ASIG-Signature header.

```
# Verify ASIG callback signature:
import hmac, hashlib

received_sig = request.headers['X-ASIG-Signature']
body_bytes = request.body
expected_sig = hmac.new(
    your_webhook_secret.encode(),
    body_bytes,
    hashlib.sha256
).hexdigest()

assert hmac.compare_digest(received_sig, expected_sig)
```

11. M1 FUND & WIRE TRANSFER RECEIVER DETAILS

For fiat wire transfers or EUR-denominated settlements, HIPERCAPITAL FINANCE must use the M1 Fund inbound pathway. This converts incoming EUR wire transfers to USDT via the ASIG tokenization engine (EUR → USD → USDT → M1 Token).

Field	Value
IBAN	AE960260001015754367301
BIC / SWIFT Code	EBILAEADXXX
Bank Name	Emirates NBD
Bank Branch	Sheikh Zayed Road, Dubai, UAE
Account Currency	USD / EUR
M1 Ingest Endpoint	POST /api/v1/payloads/m1-fund
M1 Receiver ETH Wallet	0xBD682cfD8382a90adfDd6745780D3D7959c4d939
Tokenization Flow	EUR Wire → FX Convert → USDT → M1 Token (Sepolia ERC-20)
M1 Token (Sepolia)	0xD999DB972BBDC2C13a9595A1474A04F5e59169a5

11.1 — M1 Fund Ingest Request Example

```
POST /api/v1/payloads/m1-fund
X-API-Key: asig_live_xxxx
Content-Type: application/json

{
  "grouped_m1_detail": {
    "total_group_value_eur": "250000.00",
    "group_reference": "HIPC-M1-2026-001",
    "settlement_date": "2026-06-09"
  },
  "sender": "HIPERCAPITAL FINANCE S.L.",
  "receiver": "ALSHUMOOKH GROUP",
  "callback_url": "https://your-server.com/webhook/asig"
}
```

12. INTEGRATION PHASES & COMPLIANCE REQUIREMENTS

PHASE 1 — Legal & Regulatory (Weeks 1–4)

Requirement	Detail
FCA Authorization	Obtain FCA authorization for UK payment activities or appoint an FCA-authorized partner
Independent Reserve Audit	Third-party audit of the 6.98 trillion EUR reserve claim — from a recognized firm (Big 4)
AML/KYC Program	Provide documentation of AML/KYC program — ASIG requires FATF-compliant counterparties
Legal Entity Verification	Submit UK Companies House cert + Spain CNMV registration + Panama incorporation docs

PHASE 2 — Technical Setup (Weeks 5–8)

Requirement	Detail
USDT Bridge Development	Deploy HCEURO → USDT bridge on Ethereum or Base. HCEURO has no external liquidity today
API Client Registration	Contact ASIG ops to create your API client record and receive API key + HMAC secret
IP Allowlisting	Provide your production egress IP range for registration in ASIG firewall
Webhook Endpoint	Deploy HTTPS endpoint to receive ASIG status callbacks (Section 10)
HMAC Integration	Implement HMAC-SHA256 request signing per specification in Section 3.3

PHASE 3 — Sepolia Testnet Integration (Weeks 9–12)

Requirement	Detail
Testnet API Key	Request Sepolia testnet credentials — all tests run against Chain ID 11155111
Send Test Transactions	Submit 10 test payloads with real Sepolia USDT tx_hash values
Status Polling Test	Confirm status progression: ALCHEMY_PENDING → ALCHEMY_VERIFIED → ON_CHAIN_CONFIRMED
Callback Test	Confirm webhook delivery to your callback_url on each status change
Error Code Test	Intentionally trigger each rejection code to verify your error handling

PHASE 4 — Production Go-Live (Weeks 13–16)

Requirement	Detail
Security Review	ASIG security team reviews your integration architecture
Load Test	Demonstrate handling of 200 req/60s without exceeding rate limits
First Production Payment	Submit first live USDT payment with real on-chain tx_hash
Reconciliation Confirmation	ASIG ops confirms first RECONCILED status payload
SLA Agreement	Sign ASIG integration SLA — 99.9% uptime, 48h support SLA

13. COMPLETE cURL & REQUEST EXAMPLES

13.1 — Get OAuth2 Token

```
curl -X POST https://api.alshumookh.com/api/v1/oauth/token \
-H 'Content-Type: application/x-www-form-urlencoded' \
-d 'grant_type=client_credentials&client_id=YOUR_ID&client_secret=YOUR_SECRET'
```

13.2 — Submit Settlement Payload (API Key + HMAC)

```
curl -X POST https://api.alshumookh.com/api/v1/payloads/ingest \
-H 'Content-Type: application/json' \
-H 'X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXX' \
-H 'Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000' \
-H 'X-HMAC-Signature: 3a7f9c2e4b8d1f...' \
-H 'X-Timestamp: 1749461234' \
-d '{
  "transaction_reference": "HIPC-2026-001",
  "tx_hash": "0xabc123def456...",
  "sender_wallet": "0xYOUR_WALLET",
  "receiver_wallet": "0xBD682cfD8382a90adfDd6745780D3D7959c4d939",
  "amount": "50000.00",
  "asset": "USDT",
  "network": "ethereum",
  "token_contract": "0xdAC17F958D2ee523a2206206994597C13D831ec7",
  "callback_url": "https://your-server.com/webhook/asig"
}'
```

13.3 — Check Payload Status

```
curl https://api.alshumookh.com/api/v1/payloads/a8f2c3d4-1234-5678-abcd-ef0123456789 \
-H 'X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXX'
```

13.4 — Get Schema Definition

```
curl https://api.alshumookh.com/api/v1/payloads/schema \
-H 'X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXX'
```

13.5 — Submit Using Alternative Field Names (HIPERCAPITAL style)

The normalization engine will auto-resolve HIPERCAPITAL's field names to ASIG canonical fields:

```
# HIPERCAPITAL can use their own field names – ASIG resolves aliases automatically
{
  "txRef": "HIPC-2026-001", // → transaction_reference
  "txHash": "0xabc123...", // → tx_hash
  "fromAddress": "0xYOUR_WALLET", // → sender_wallet
  "toAddress": "0xBD682c...", // → receiver_wallet
  "tokenAmount": "50000.00", // → amount
  "tokenSymbol": "USDT", // → asset
  "chainName": "ethereum", // → network
  "contractAddress": "0xdAC17F...", // → token_contract
  "webhookUrl": "https://...", // → callback_url
  "authorizationCode": "HIPC-AUTH-001" // → authorization_code
}
```

14. COMPLIANCE CHECKLIST & GO-LIVE CRITERIA

ALL items in this checklist must be completed and verified by ALSHUMOOKH GROUP before HIPERCAPITAL FINANCE will be approved for production access to ASIG.

Category	Requirement	Evidence Required	Status
Legal	FCA Authorization (UK) or equivalent EU/UAE financial license	License certificate + FCA register entry	PENDING
Legal	AML/KYC Program documentation	AML policy document + compliance officer contact	PENDING
Legal	Corporate structure chart + beneficial ownership	Legal entity chart + UBO declaration	PENDING
Financial	Independent reserve audit for HCEURO backing claim	Audit report from Big 4 or equivalent firm	PENDING
Financial	HCEURO → USDT bridge operational on public chain	Bridge contract address + audit + liquidity proof	PENDING
Technical	USDT test transactions on Sepolia (10 successful payloads)	ASIG payload IDs with RECONCILED status	PENDING
Technical	HMAC-SHA256 signing implemented and verified	Demo request with valid HMAC header	PENDING
Technical	Webhook endpoint deployed and tested	All status transitions received successfully	PENDING
Technical	Idempotency-Key implementation verified	Duplicate request correctly returns replay response	PENDING
Technical	Rate limiting compliance (200 req/60s)	Load test results showing no rate limit violations	PENDING
Operational	IP allowlist registered in ASIG	Production egress IPs provided to ASIG ops	PENDING
Operational	SLA agreement signed	Signed SLA document	PENDING

■ **CRITICAL BLOCKER:** HIPERCAPITAL FINANCE cannot go live on ASIG without completing Item 5 (HCEURO → USDT bridge). The ASIG system does not accept HCEURO, TOKENIZED_USD, USDX, or any private ledger asset. All settlements must use publicly verifiable USDT on Ethereum, Base, or Tron with a real on-chain tx_hash.

For integration support or to begin the onboarding process, contact ALSHUMOOKH GROUP Technical Operations. Reference: ASIG-HIPC-TECH-2026-v3

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